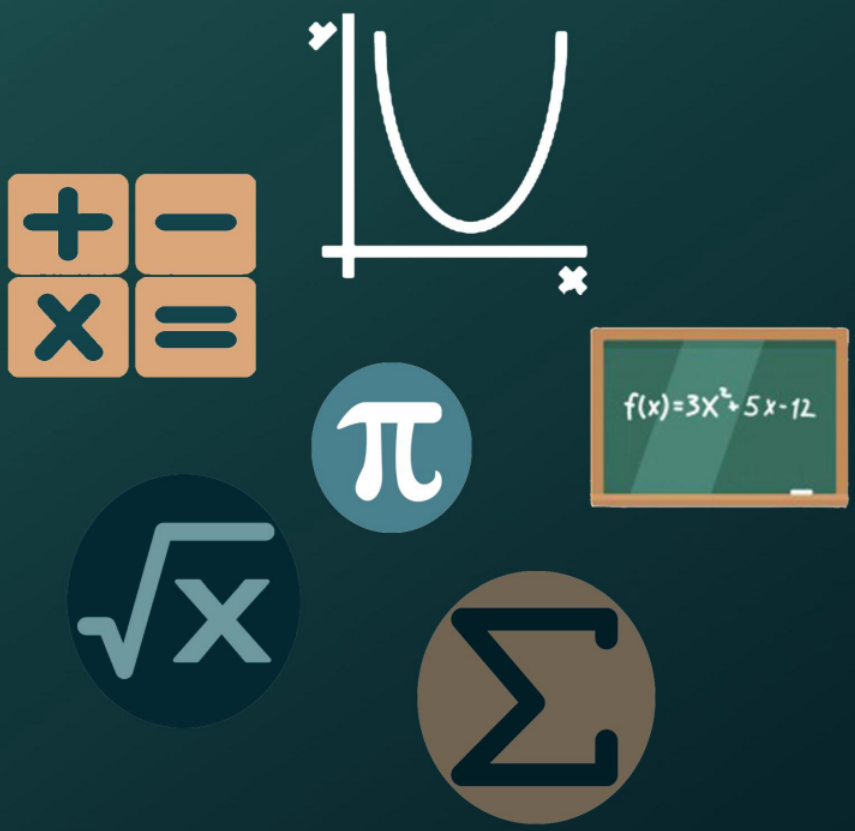
MATHEMATICS

Complex Number



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Complex Number	1-3	1		
2	Algebra of Complex Number	4-7	2,23	1	1
3	Conjugate of Complex Number	8-11	9-13	2	
4	Modulus of Complex Number	12-18	19		
5	Argument of Complex Number	19-21		3-5	
6	Properties of Amplitude	22,23	3, 24-26	6	31
7	Representation of Complex Number: Cartesian & Vector Form	24	27	7	
8	Representation of Complex Number: Trigonometric/Polar Form	25,26		8,9	
9	Representation of Complex Number: Exponential Form	27,28	28		
10	De-Moivre's Theorem	29			
11	Cube Root of Unity	30-35	4,5,29	10-17	2
12	nth Roots of Unity	36	14-16,20	18,19	3
13	Geometry of Complex Number : Point	37-39	7	20-22	4
14	Rotation of Complex Number		30,31	23	32,33
15	Geometry of Complex Number : Straight Line		8,17,18		5,6
16	Geometry of Complex Number : Conic Section	40	6	24	7-10, 15-20
17	Locus of a Complex Number	41,42	32	25-27	11-12
18	Triangular Inequality	43-46		28,29	13, 21-24
19	After Complete Chapter		21,22		14, 25-30, 34-37

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Complex Number

Q1. The value of $1+i^2+i^4+i^6+.....+i^{2n}$ is
(A) Positive
(B) Negative
(C) Zero
(D) non negative

Q2. $\frac{i^{4n+3}+(-i)^{8n-3}}{(-i)^{12n-1}-i^{2-16n}}$, $n \in N$ is equal to -
(A) 1 + i (B) 2i
(C) –2i (D) –1 – i

Q3. If $i^2 = -1$, then the value of $\sum_{n=1}^{200} i^n$ is-
(A) 50 (B) –50
(C) 0 (D) 100

Algebra of Complex Number

Q4. If $\left(\frac{1-i}{1+i}\right)^{100} = a+ib$, then :
(A) a = 0, b = 1 (B) a = 1, b = 0
(C) a = 2, b = –1 (D) a = –1, b = 2

Q5. If (x + iy) (2 –3i) = 4 + i, then
(A) x = – 14/3, y = 5/13
(B) x = 5/13. y = 14/13
(C) x = 14/13, y = 5/13
(D) x = 5/13, y = –14/13

Q6. The square root of 3 + 4i is -
(A) $\pm(2-i)$ (B) $\pm(1-2i)$
(C) $\pm(2+i)$ (D) $\pm(1+2i)$

Q7. If $(2+i\sqrt{3})$ is a root of the equation $x^2+px+q=0$, where p and q are real, then (p, q) equals to:
(A) (–4, 7) (B) (4, –7)
(C) (–4, 7) (D) (4, 7)

Conjugate of Complex Number

Q8. The conjugate of $\frac{1}{3+4i}$ is-
(A) (3 – 4i) (B) $\frac{1}{25}(3+4i)$
(C) $\frac{1}{25}(3-4i)$ (D) None of these

Q9. The conjugate of the complex number $\frac{2+5i}{4-3i}$ is -
(A) $\frac{-7-26i}{25}$ (B) $\frac{7-26i}{25}$
(C) $\frac{7+26i}{25}$ (D) $\frac{-7+26i}{25}$

Q10. If the conjugate of $(x+iy)(1-2i)$ be 1+ i, then
(A) $y=\frac{3}{5}$ (B) $x-iy=\frac{1-i}{1+2i}$
(C) $x+iy=\frac{1-i}{1-2i}$ (D) $x=\frac{1}{5}$

Q11. If z = x + iy then the value of $\frac{z-\bar{z}}{z+\bar{z}}$ is -
(A) $\frac{y}{ix}$ (B) $-\frac{ix}{y}$
(C) $\frac{ix}{y}$ (D) $\frac{iy}{x}$

Modulus of Complex Number

Q12. If z_1 and z_2 are any two complex numbers, then $|z_1+z_2|^2+|z_1-z_2|^2$ is equal to :

(A) $2|z_1|^2|z_2|^2$ (B) $2|z_1|^2+2|z_2|^2$

(C) $|z_1|^2+|z_2|^2$ (D) $2|z_1||z_2|$

Q13. $\left|(1+i)\frac{(2+i)}{(3+i)}\right|$ is equal to

(A) 1 (B) –1

(C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

Q14. If $z=\frac{(3+4i)(5-7i)}{(7+5i)(4-3i)}$ then $|z|$ = ?

(A) 5 (B) 4

(C) 1 (D) None of these

Q15. If $z_1, z_2, \in C$ are such that

$|z_1+z_2|^2=|z_1|^2+|z_2|^2$ then $\frac{z_1}{z_2}$ is -

(A) Zero

(B) Purely real

(C) Purely imaginary

(D) Complex

Q16. If $\frac{2z_1}{3z_2}$ is a purely imaginary number, then $\left|\frac{z_1-z_2}{z_1+z_2}\right|$ is equal to :

(A) $\frac{3}{2}$ (B) 1 (C) $\frac{2}{3}$ (D) $\frac{4}{9}$

Q17. If the number $\frac{z-1}{z+1}$ is purely imaginary, then

(A) $|z|>1$ (B) $|z|=1$

(C) $|z|<1$ (D) None of these

Q18. The solution of the equation $|z|-z=1+2i$ is

(A) $2-\frac{3}{2}i$ (B) $\frac{3}{2}+2i$

(C) $\frac{3}{2}-2i$ (D) $-2+\frac{3}{2}i$

Argument of Complex Number

Q19. The argument of the complex number $-1+i\sqrt{3}$ is

(A) 120° (B) –120°

(C) –60° (D) 60°

Q20. If $\arg(z)=\theta$, then $\arg(\bar{z})$ is equal to

(A) $\theta-\pi$ (B) $\pi-\theta$

(C) θ (D) $-\theta$

Q21. The argument of the complex number $\frac{13-5i}{4-9i}$ is -

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{5}$ (D) $\frac{\pi}{6}$

Properties of Amplitude

Q22. The amplitude of $\sin\frac{\pi}{5}+i\left(1-\cos\frac{\pi}{5}\right)$

(A) $\frac{\pi}{5}$ (B) $\frac{2\pi}{5}$

(C) $\frac{\pi}{10}$ (D) $\frac{\pi}{15}$

Q23. The amplitude of $\frac{a+ib}{a-ib}$ is -

(A) $\tan^{-1}\left(\frac{2ab}{a^2+b^2}\right)$ (B) $\tan^{-1}\left(\frac{2ab}{b^2+a^2}\right)$

(C) $\tan^{-1}\left(\frac{2ab}{a^2-b^2}\right)$ (D) $\tan^{-1}\left(\frac{a^2-b^2}{2ab}\right)$

Representation of Complex Number: Cartesian & Vector Form

Q24. If $z = -3 + 2i$, then $1/z$ is equal to

(A) $\frac{1}{13}(3 + 2i)$ (B) $-\frac{1}{13}(3 + 2i)$

(C) $\frac{1}{\sqrt{13}}(3 + 2i)$ (D) $-\frac{1}{\sqrt{13}}(3 + 2i)$

Representation of Complex Number: Trigonometric/Polar Form

Q25. Let z be a complex number such that $|z| = 4$ and $\arg(z) = \frac{5\pi}{6}$, then z equals-

(A) $-2\sqrt{3} + 2i$ (B) $2\sqrt{3} + 2i$

(C) $2\sqrt{3} - 2i$ (D) $-\sqrt{3} + i$

Q26. The polar form of $(-1 + i)$ is-

(A) $\sqrt{2}(\cos\pi/4 + i\sin\pi/4)$

(B) $\sqrt{2}(\cos5\pi/4 + i\sin5\pi/4)$

(C) $\sqrt{2}(\cos3\pi/4 + i\sin3\pi/4)$

(D) $\sqrt{2}(\cos\pi/4 - i\sin\pi/4)$

Representation of Complex Number: Exponential Form

Q27. $-1+\sqrt{-3} = re^{i\theta}$, then θ is equal to -

(A) $\frac{2\pi}{3}$ (B) $-\frac{\pi}{3}$

(C) π (D) $-\frac{2\pi}{3}$

Q28. $|z_1 + z_2| = |z_1| + |z_2|$ is possible, if

(A) $z_2 = z_1$ (B) $z_2 = \frac{1}{z_1}$

(C) are $(z_1) = \arg(z_2)$ (D) $|z_1| = |z_2|$

De-Moivre's Theorem

Q29. The value of

$$\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\left(\cos\left(\frac{\pi}{2^2}\right) + i\sin\left(\frac{\pi}{2^2}\right)\right)$$
$$\left(\cos\left(\frac{\pi}{2^3}\right) + i\sin\left(\frac{\pi}{2^3}\right)\right).....\infty$$

is :

(A) –1 (B) 1

(C) 0 (D) $\sqrt{2}$

Cube Root of Unity

Q30. One of the cube roots of unity is :

(A) $\frac{1-i\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}-i}{2}$

(C) $\frac{-1+i\sqrt{3}}{2}$ (D) $\frac{1+i\sqrt{3}}{2}$

Q31. If 1, ω, ω^2 are the three cube roots of unity, then $(3+\omega^2+\omega^4)^6$ is

(A) 0 (B) 2

(C) 64 (D) 729

Q32. The three value of $(8)^{1/3}$ are -

(A) 2, 2, 2 (B) 2, 2ω, $2\omega^2$

(C) –2, -2ω, $-2\omega^2$ (D) 2, ω, ω^2

Q33. If α is an imaginary cube root of unity, then for $n \in N$, the value of $\alpha^{3n+1} + \alpha^{3n+3} + \alpha^{3n+5}$.

(A) –1 (B) 0

(C) 3 (D) 1

Q34. If α and β are imaginary cube roots of unity, then the value of $\alpha^4 + \beta^{28} + \frac{1}{\alpha\beta}$ is :

(A) 1 (B) –1

(C) 0 (D) None of these

Q35. If ω is the cube root of unity, then $(3+5\omega+3\omega^2)^2+(3+3\omega+5\omega^2)^2$ is equal to -

(A) 4 (B) –4

(C) 0 (D) None of these

n^{th} Roots of Unity

Q36. The nth roots of unity are in :

(A) AP (B) GP

(C) HP (D) None of these

Geometry of Complex Number : Point

Q37. If $z = x + iy$, then area of the triangle whose vertices are points z, iz, z + iz is :

(A) $\frac{1}{2}|z|^2$ (B) $\frac{1}{4}|z|^2$

(C) $|z|^2$ (D) $\frac{3}{2}|z|^2$

Q38. In which quadrant of the complex plane, the point $\frac{1+2i}{1-i}$ lies?

(A) Fourth (B) First

(C) Second (D) Third

Q39. Let z_1 and z_2 be two complex number such that $\frac{z_1}{z_2}+\frac{z_2}{z_1}=1$, then

(A) z_1, z_2 are collinear

(B) z_1, z_2 and the origin form a right angled triangle

(C) z_1, z_2 and the origin form an equilateral triangle

(D) none of these

Geometry of Complex Number : Conic Section

Q40. The equation $z\bar{z}+(2-3i)z+(2+3i)\bar{z}+4=0$ represents a circle of radius.

(A) 2 (B) 3

(C) 4 (D) 6

Locus of a Complex Number

Q41. If $\left|\frac{z-i}{z+i}\right|=1$, then locus of z is

(A) x-axis (B) y-axis

(C) x = 1 (D) y = 1

Q42. A complex number z is such that $\arg\left(\frac{z-2}{z+2}\right)=\frac{\pi}{3}$. The points representing this complex number will lie on :

(A) A straight line (B) An ellipse

(C) A parabola (D) A circle

Triangular Inequality

Q43. If z_1 and z_2 are two complex numbers, then $|z_1 - z_2|$

(A) $\geq |z_1| - |z_2|$ (B) $\geq |z_1| + |z_2|$

(C) $\leq |z_1| + |z_2|$ (D) $\leq |z_1| - |z_2|$

Q44. If z_1, z_2 are any two complex numbers, then:

(A) $|z_1+z_2| \geq |z_1|+|z_2|$

(B) $|z_1+z_2| > |z_1|+|z_2|$

(C) $|z_1+z_2| \leq |z_1|+|z_2|$

(D) $|z_1+z_2| = |z_1|+|z_2|$

Q45. If $|z| \leq 1$ then the minimum and maximum value of $|z - 3|$ are -

(A) 4, 2 (B) 3, 4

(C) 4, 6 (D) 2, 4

Q46. For any complex number z, the minimum value of $|z| + |z - 1|$ is

(A) 0 (B) 1

(C) 1/2 (D) 3/2

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The value of $i^{57} + 1/i^{125}$ is not equal to :-

(A) 0 (B) $-2i$

(C) $2i$ (D) 2

Q2. If x and y are satisfying the equation

$$\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y+i}{3-i}=i \text{ then-}$$

(A) $x+y>0$ (B) $x+y>1$

(C) $x+y>2$ (D) $x+y>3$

Q3. For two complex numbers z_1 and z_2 :

$(az_1+b\bar{z}_1)(cz_2+d\bar{z}_2)=(cz_1+d\bar{z}_1)(az_2+b\bar{z}_2)$

if (a, b, c, d $\in$ R)-

(A) $\frac{a}{b}=\frac{c}{d}$ (B) $\frac{a}{d}=\frac{b}{c}$

(C) $|z_1|=|z_2|$ (D) $\arg(z_1)=\arg(z_2)$

Q4. Let z and ω are two complex numbers such that $|z|\le 1$, $|\omega|\le 1$ and $|z+i\omega|=|z-i\bar{\omega}|=2$, then $z+\omega$ can be equal to -

(A) $1+i$ (B) $1-i$

(C) $-1+i$ (D) $-1-i$

Q5. If α & β are imaginary cube roots of unity then $\alpha^n+\beta^n$ is equal to (where n $\in$ I) -

(A) $2\cos\frac{2n\pi}{3}$ (B) $\cos\frac{2n\pi}{3}$

(C) $2i\sin\frac{2n\pi}{3}$ (D) $i\sin\frac{2n\pi}{3}$

Q6. The locus of the complex number z in argand plane satisfying the inequality

$$\log_{1/2}\left(\frac{|z-1|+4}{3|z-1|-2}\right)>1 \left(\text{where } |z-1|\neq\frac{2}{3}\right)$$

is -

(A) A circle

(B) Interior of a circle

(C) Exterior of a circle

(D) None of these

Q7. If $\left|z-\frac{4}{z}\right|=2$, then the greatest value of $|z|$ is -

(A) $1+\sqrt{2}$ (B) $2+\sqrt{2}$

(C) $\sqrt{3}+1$ (D) $\sqrt{5}+1$

Q8. Which of the following, locii of z on the complex plane represents a pair of straight lines ?

(A) $\text{Re}(z^2)=0$ (B) $\text{Im}(z^2)=0$

(C) $|z|+z=0$ (D) $|z-1|=|z-i|$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 9 to 13)

If conjugate of a complex number $z = x + iy$ is defined as $\bar{z}=x-iy$, then -

Q9. The value of $\overline{z_1+z_2}$ is equal to-

(A) $\bar{z}_1+\bar{z}_2$ (B) $\bar{z}_1-\bar{z}_2$

(C) $z_1-\bar{z}_2$ (D) None of these

Q10. $\overline{z_1 z_2}$ is equal to-

(A) $\overline{z_1}\,\overline{z_2}$ (B) $i\overline{z_1}\,\overline{z_2}$

(C) $-i\overline{z_1}\,\overline{z_2}$ (D) None of these

Q11. $\overline{\left(\frac{z_1}{z_2}\right)}$ is equal to-

(A) $\frac{\overline{z_1}}{z_2}$ (B) $i\,\frac{\overline{z_1}}{z_2}$

(C) $-i\,\frac{\overline{z_1}}{z_2}$ (D) None of these

Q12. $\overline{z+\bar{z}}$ is equal to-

(A) $2z$ (B) $\bar{z}+z$

(C) $\bar{z}+iz$ (D) None of these

Q13. $\overline{\left(\frac{z_1 z_2}{z_3}\right)}$ is equal to-

(A) $i\,\frac{\overline{z_1}\,\overline{z_2}}{\overline{z_3}}$ (B) $\frac{\overline{z_1}\,\overline{z_2}}{\overline{z_3}}$

(C) $\frac{z_1 z_2}{z_3}$ (D) None of these

Paragraph # 2 (Ques. 14 to 16)

Let z be any complex number. To factorise the expression of the form $z^n - 1$, we consider the equation $z^n = 1$. This equation is solved using De moiver's theorem.

Let $1, \alpha_1, \alpha_2, \ldots\ldots \alpha_{n-1}$ be the roots of this equation, then $z^n - 1 = (z-1)(z-\alpha_1)(z-\alpha_2)\ldots\ldots(z-\alpha_{n-1})$ This method can be generalised to factorize any expression of the form $z^n - k^n$.

For example, $z^7 + 1 = \prod_{m=0}^{6}\left(z - \text{CiS}\left(\frac{2m\pi}{7}+\frac{\pi}{7}\right)\right)$

This can be further simplified as

$z^7 + 1 = (z+1)$

$\left(z^2 - 2z\cos\frac{\pi}{7}+1\right)\left(z^2 - 2z\cos\frac{3\pi}{7}+1\right)$

$\left(z^2 - 2z\cos\frac{5\pi}{7}+1\right)$ ..... (i)

These factorisations are useful in proving different trigonometric identities e.g. in equation (i) if we put $z = i$, then equation (i) becomes

$$(1-i) = (i+1)\left(-2i\cos\frac{\pi}{7}\right)\left(-2i\cos\frac{3\pi}{7}\right)\left(-2i\cos\frac{5\pi}{7}\right)$$

i.e. $\cos\frac{\pi}{7}\cos\frac{3\pi}{7}\cos\frac{5\pi}{7} = -\frac{1}{8}$

Q14. If the expression $z^5 - 32$ can be factorised into linear and quadratic factors over real coefficients as

$(z^5 - 32) = (z-2)(z^2 - pz + 4)(z^2 - qz + 4)$,

where $p > q$, then the value of $p^2 - 2q$-

(A) 8 (B) 4

(C) –4 (D) –8

Q15. By using the factorisation for $z^5 + 1$, the value of $4\sin\frac{\pi}{10}\cos\frac{\pi}{5}$ comes out to be -

(A) 4 (B) 1/4

(C) 1 (D) –1

Q16. If $(z^{2n+1} - 1) = (z-1)(z^2 - p_1 z + 1)\ldots\ldots (z^2 - p_n z + 1)$ where $n \in N$ & $p_1, p_2 \ldots\ldots p_n$ are real numbers then $p_1 + p_2 + \ldots\ldots + p_n =$

(A) –1 (B) 0

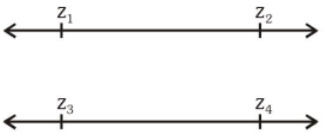
(C) $\tan(\pi/2n)$ (D) None of these

MATCH THE COLUMN TYPE

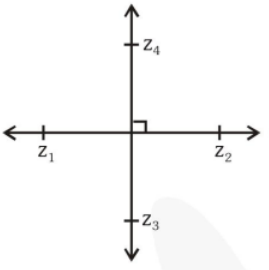
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. Match the figure in column-I with corresponding expression -

Column-I **Column - II**

(A) 

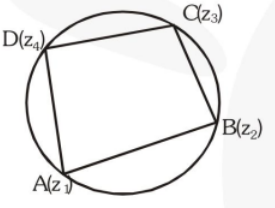
two parallel lines (P) $\frac{z_4 - z_3}{z_2 - z_1} + \frac{\bar{z}_4 - \bar{z}_3}{\bar{z}_2 - \bar{z}_1} = 0$

(B) 

two perpendicular lines (Q) $\frac{z_2 - z_1}{z_4 - z_3} = \frac{\bar{z}_2 - \bar{z}_1}{\bar{z}_4 - \bar{z}_3}$

(C) 

a parallelogram (R) $\frac{z_4 - z_1}{z_2 - z_1} \cdot \frac{z_2 - z_3}{z_4 - z_3} = \frac{\bar{z}_4 - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} \cdot \frac{\bar{z}_2 - \bar{z}_3}{\bar{z}_4 - \bar{z}_3}$

(D) 

(S) $z_1 + z_3 = z_2 + z_4$

Q18. **Column - I** **Column - II**

(A) If G be the greatest and L be the least values of $|z - 1|$, if $|z + 2 + i| \le 1$ where $i = \sqrt{-1}$, then (P) LG = 9

(B) If G be the greatest and L be the least values of $|z + 2i|$, where $i = \sqrt{-1}$, if $1 \le |z - 1| \le 3$, then (Q) L + G = 6

(C) If g be the greatest and L be the least values of $|z - 2|$, if $|z + i| \le 1$ where $i = \sqrt{-1}$, then (R) $\left(\sqrt{2G} - \sqrt{2L}\right)^2 = 4$

(S) LG = 4

(T) G – L = 2

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. z_1 and z_2 are two complex numbers such that $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unimodular (whose modulus is one), while z_2 is not unimodular. Find $|z_1|$.

Q20. Find the value $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - \cos\frac{2\pi k}{7}\right)$

Q21. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) has 4 non real roots, two with sum 3 + 4i and the other two with product 13 + i. Find the value of 'b'.

Q22. Let $P(z) = z^3 + az^2 + bz + c$, where a, b and c are real. There exist a complex number w such that the three roots of P(z) = 0, are w + 3 i, w + 9 i and 2w – 4 where $i^2 = -1$ then |a + b + c| is equal to.

SUBJECTIVE TYPE

Q23. If $x = -5 + 2\sqrt{-4}$, find the value of $x^4 + 9x^3 + 35x^2 - x + 4$.

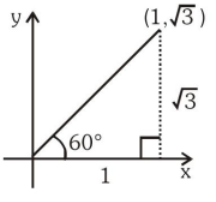
Q24. Find the modulus, argument, principal value of argument, least positive argument of complex numbers.

(A) $1 + i\sqrt{3}$

(B) $-1 + i\sqrt{3}$

(C) $1 - i\sqrt{3}$

(D) $-1 - i\sqrt{3}$



Q25. Find modulus and argument for $z = 1 - \sin\alpha + i\cos\alpha$, $\alpha \in (0, 2\pi)$

Q26. Find amp z and |z| if $z = \left[\frac{(3+4i)(1+i)(1+\sqrt{3}i)}{(1-i)(4-3i)(2i)}\right]^2$.

Q27. Among the complex number z which satisfies $|z - 25i| \le 15$, find the complex numbers z having

(A) least positive argument

(B) maximum positive argument

(C) least modulus

(D) maximum modulus

Q28. Express the following complex numbers in polar and exponential form :

(i) $\frac{1+3i}{1-2i}$ (ii) $\frac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$

Q29. If α, β, γ are roots of $x^3 - 3x^2 + 3x + 7 = 0$ (and ω is imaginary cube root of unity), then find the value of $\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1}$.

Q30. Complex numbers z_1, z_2, z_3 are the vertices A, B, C respectively of an isosceles right angled triangle with right angle at C. Show that $(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2)$.

Q31. If the vertices of a square ABCD are z_1, z_2, z_3 & z_4 then find z_3 & z_4 in terms of z_1 & z_2.

Q32. Find the locus of :

(A) $|z - 1|^2 + |z + 1|^2 = 4$

(B) $Re(z^2) = 0$

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let z is complex number satisfying the equation $z^2-(3+i)z+m+2i=0$, where $m \in R$. Suppose the equation has a real root. The additive inverse of non real root, is

(A) $1-i$ (B) $1+i$

(C) $-1-i$ (D) -2

Q2. Number of complex numbers z satisfying $z^3 = \bar{z}$ is

(A) 1 (B) 2

(C) 4 (D) 5

Q3. Let z be a complex number satisfying the equation $(z^3+3)^2 = -16$ then $|z|$ has the value equal to

(A) $5^{1/2}$ (B) $5^{1/3}$ (C) $5^{2/3}$ (D) 5

Q4. If $w=\alpha+i\beta$ where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\frac{w-\bar{w}z}{1-z}$ is purely real, then the set of values of z is -

(A) $\{z : |z|=1\}$

(B) $\{z : z=\bar{z}\}$

(C) $\{z : z \neq 1\}$

(D) $\{z : |z| = 1, z \neq 1\}$

Q5. Consider $az^2+bz+c=0$, where a, b, c $\in$ R and $4ac > b^2$.

If z_1 and z_2 are the roots of the equation given above, then which of the following complex numbers is purely real ?

(A) $z_1\bar{z}_2$ (B) $\bar{z}_1z_2$

(C) z_1-z_2 (D) $(z_1-z_2)i$

Q6. If $z=\frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right)$,

then $\left(\frac{|z|}{\text{amp } z}\right)$ equals -

(A) 1 (B) π

(C) 3π (D) 4

Q7. If

$|z-i\,\text{Re}(z)| = |z-\text{Im}(z)|, (\text{where } i=\sqrt{-1})$,

then z lies on -

(A) Re (z) = 2

(B) Im (z) = 2

(C) Re (z) + Im (z) =2

(D) None of these

Q8. z is a complex number such that $z+\frac{1}{z} = 2\cos 3°$. Then the value of $z^{2000}+\frac{1}{z^{2000}}+1$ is equal to

(A) 0 (B) -1

(C) $\sqrt{3}+1$ (D) $1-\sqrt{3}$

Q9. If $\cos\alpha + \cos\beta + \cos\gamma = 0 = \sin\alpha + \sin\beta + \sin\gamma$, then $\sin 3\alpha + \sin 3\beta + \sin 3\gamma$ is equal to -

(A) $\sin 3(\alpha+\beta+\gamma)$

(B) $3\sin(\alpha+\beta+\gamma)$

(C) $3\sin 3(\alpha+\beta+\gamma)$

(D) $3\cos(\alpha+\beta+\gamma)$

Q10. If 1, ω and ω^2 are the three cube roots of unity, then the roots of the equation $(x-1)^3-8=0$ are

(A) $-1, -1, -2\omega, -1+2\omega^2$

(B) $3, 2w, 2w^2$

(C) $3, 1+2\omega, 1+3\omega^2$

(D) None of these

Q11. When the polynomial $5x^3 + Mx + N$ is divide by $x^2 + x + 1$ the remainder is 0. The value of (M + N) is equal to -

(A) – 3 (B) 5

(C) – 5 (D) 15

Q12. If the expression $x^{2m} + x^m + 1$ is divisible by $x^2 + x + 1$, then :

(A) m is any odd integer

(B) m is divisible by 3

(C) m is not divisible by 3

(D) None of these

Q13. If $z^2 - z + 1 = 0$, then the value of

$$\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2 + ... + \left(z^{24} + \frac{1}{z^{24}}\right)^2$$ is equal to

(A) 24 (B) 32

(C) 48 (D) None of these

Q14. If 1, ω, ω^2 are the cube roots of unity, then $(1 + \omega - \omega^2)(1 - \omega + \omega^2)$ equals :

(A) –4 (B) –2

(C) 2 (D) 4

Q15. If $z = \left(\frac{\sqrt{3} + i}{2}\right)^5 + \left(\frac{\sqrt{3} - i}{2}\right)^5$, then

(A) Im (z) = 0

(B) Re (z) = 0

(C) Re (z) > 0, Im (z) < 0

(D) Re (z) > 0, Im (z) > 0

Q16. If a, b, c are integers not all equal and ω is cube root of unity ($\omega \neq 1$) then the minimum value of $|a + b\omega + c\omega^2|$ is -

(A) 0 (B) 1

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

Q17. $\left(\sqrt[3]{3} + \left(3^{5/6}\right)i\right)^3$ is an integer where $i = \sqrt{-1}$.

The value of the integer is equal to

(A) 24 (B) –24

(C) –22 (D) –21

Q18. Let α be the root of the equation $z^5 - 1 = 0$ such that $\alpha \neq 1$. Then the value of $\sum_{r=15}^{50} \alpha^r$ is equal to

(A) 1 (B) i

(C) – 1 (D) 0

Q19. If 1, ω, ω^2, ..., ω^{n-1} are nth roots of unity, the value of $(9 - \omega)(9 - \omega^2)$ ..... $(9 - \omega^{n-1})$ will be

(A) n (B) 0

(C) $\frac{9^n - 1}{8}$ (D) $\frac{9^n + 1}{8}$

Q20. The figure formed by four points 1 + 0 i ; –1 + 0 i ; 3 + 4 i & $\frac{25}{-3-4i}$ on the argand plane is :

(A) a parallelogram but not a rectangle

(B) a trapezium which is not equilateral

(C) a cyclic quadrilateral

(D) None of these

Q21. The number of points in the complex plane that satisfying the conditions $|z - 2| = 2$, $z(1 - i) + \bar{z}(1 + i) = 4$, $\left(\text{where } i = \sqrt{-1}\right)$ is

(A) 0

(B) 1

(C) 2

(D) More than 2

Q22. If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on :

(A) a line not passing through the origin

(B) $|z| = \sqrt{2}$

(C) the x-axis

(D) the y-axis

Q23. A man walks a distance of 3 units from the origin towards the north-east (N 45° E) direction. From there, he walks a distance of 4 units towards the north-west (N 45° W) direction to reach a point P. Then the position of P in the Argand plane is :

(A) $3e^{i\pi/4} + 4i$ (B) $(3 - 4i)e^{i\pi/4}$

(C) $(4 + 3i)e^{i\pi/4}$ (D) $(3 + 4i)e^{i\pi/4}$

Q24. If z be complex number such that equation $|z-a^2|+|z-2a|=3$ always represents an ellise, then range of a $(\in R^+)$ is

(A) $(1,\sqrt{2})$ (B) $\left[1,\sqrt{3}\right]$

(C) $(-1,3)$ (D) $(0,3)$

Q25. On the complex plane locus of a point z satisfying the inequality $2 \leq |z - 1| < 3$ denotes

(A) region between the concentric circles of radii 3 and 1 centered at (1, 0)

(B) region between the concentric circles of radii 3 and 2 centered at (1, 0) excluding the inner and outer boundaries.

(C) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner and outer boundaries.

(D) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner boundary and excluding the outer boundary.

Q26. If the following regions in the complex plane, the only one that does not represent a circle is -

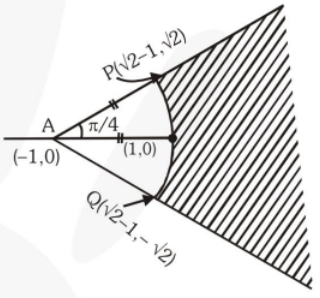
(A) $z\bar{z} + i(z - \bar{z}) = 0$

(B) $\text{Re}\left(\frac{1+z}{1-z}\right) = 0$

(C) $\arg\left(\frac{z-i}{z+i}\right) = \frac{\pi}{2}$

(D) $\left|\frac{z-i}{z+i}\right| = 1$

Q27. Area of shaded region belongs to -



(A) $z : |z + 1| > 2, |\arg(z + 1)| < \pi/4$

(B) $z : |z - 1| > 2, |\arg(z - 1)| < \pi/4$

(C) $z : |z + 1| < 2, |\arg(z + 1)| < \pi/2$

(D) $z : |z - 1| < 2, |\arg(z - 1)| < \pi/2$

Q28. If $|z - i| \leq 2$ and $z_1 = 5 + 3i$, $\left(\text{where } i = \sqrt{-1}\right)$ then the maximum value of $|iz + z_1|$ is -

(A) $2 + \sqrt{31}$ (B) 7

(C) $\sqrt{31} - 2$ (D) $2 + \sqrt{31}$

Q29. If the complex number z satisfies the condition $|z| \geq 3$, then the least value of $\left|z + \frac{1}{z}\right|$ is equal to:

(A) 5/3 (B) 8/3

(C) 11/3 (D) None of these

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The equation $z^2 - i|z - 1|^2 = 0$ has
(A) no real root
(B) no purely imaginary root
(C) all roots inside $|z| < 1$
(D) at least two roots

Q2. The common roots of the equations
$z^3 + (1 + i)z^2 + (1 + i)z + i = 0$,
(where $i = \sqrt{-1}$) and $z^{1993} + z^{1994} + 1 = 0$ are
(A) 1 (B) ω (C) ω^2 (D) ω^{981}

Q3. If $\alpha_1, \alpha_2, \alpha_3, \ldots\ldots, \alpha_{n-1}$ are the imaginary n^{th} roots of unity then the product $\prod_{r=1}^{n-1}(i-\alpha_r)$.
(where $i = \sqrt{-1}$) can take the value equal to
(A) 0 (B) 1
(C) i (D) (1 + i)

Q4. Let z, ωz and $z + \omega z$ represent three vertices of ΔABC, where ω is cube root unity, then -
(A) centroid of ΔABC is $\frac{2}{3}(z + \omega z)$
(B) orthocenter of ΔABC is $\frac{2}{3}(z + \omega z)$
(C) ABC is an obtuse angled triangle
(D) ABC is an acute angled triangle

Q5. If O is the origin and Oz_1 and Oz_2 are perpendicular then one must have
(A) $z_1 = iz_2$
(B) $|z_1 - z_2| = |z_1 + z_2|$
(C) $\text{Re}(z_1\overline{z}_2 + z_2\overline{z}_1) = 0$
(D) $\text{Re}\left(\frac{z_2}{z_1}\right) = 0$

Q6. If z_1 and z_2 are complex numbers scuh that $\text{Re}\left(\frac{z_2}{z_1}\right) = 0$ then
(A) Oz_1 and Oz_2 are perpendicular, where O is the origin
(B) $|z_1 - z_2| = |z_1 + z_2|$
(C) $z_1 = \pm i z_2$
(D) $z_1\overline{z}_2 + z_2\overline{z}_1 = 0$

Q7. If the complex number z satisfies $|iz + 2| = \text{Im}(z)$, then
(A) minimum value of $|z|$ is 1
(B) maximum value of $|z|$ is $\sqrt{5}$
(C) minimum value of arg z is $\frac{\pi}{4}$
(D) maximum value of arg z is $\frac{3\pi}{4}$

Q8. z_1, z_2 lie on the unit circle $|z| = 1$ and satisfy $z_1\overline{z}_2 = -1$, then one has
(A) $z_1 + z_2 = 0$ (B) $z_1^2 + z_2^2 = 0$
(C) $z_1^3 + z_2^3 = 0$ (D) $\text{Im}\left(\frac{z_1}{z_2}\right) = 0$

Q9. If $\frac{z+1}{z+i}$ is a purely imaginary number,
(where $i = \sqrt{-1}$) then z lies on a
(A) straight line
(B) circle
(C) circle with radius $= 1/\sqrt{2}$
(D) circle passing through the origin

Q10. The points z satisfying $|z - 1 + 3i| = 2|z - 1|$ also satisfy
(A) $|z+1-i| = 2$
(B) $z\overline{z} + (1+i)\overline{z} + (1-i)z = 2$
(C) $\arg\frac{z-i}{z-3-i} = \frac{\pi}{2}$
(D) $|z-1-i| = 2$

11. If z is a complex number satisfying the equation $|z-1|=1$ and $\omega=\frac{3}{z}$, then locus of $\omega = x + iy$ in complex plane is -

(A) $2x-3=0$ (B) $2x+3=0$

(C) $2x-3y=0$ (D) $2x+3y=0$

Q12. On the argand plane, let complex number α & β satisfing $\alpha=-2+3z$, $\beta=-2-3z$ & $|z|=1$. Then the correct statement is -

(A) α moves on the circle, centre at (–2,0) and radius 3

(B) α & β describe the same locus

(C) α & β move on different circles

(D) $\alpha-\beta$ moves on a circle concentric with $|z|=1$

Q13. If $z = x + iy$, then

(A) $|z|\le|\mathrm{Re}\,z|+|\mathrm{Im}\,z|$

(B) $|\mathrm{Re}\,z|+|\mathrm{Im}\,z|\le\sqrt{2}|z|$

(C) $|\mathrm{Re}\,z||\mathrm{Im}\,z|\le z$

(D) $\mathrm{Re}\,z+|\mathrm{Im}\,z|\le 2|z|$

Q14. Let z_1, z_2, z_3 be non-zero complex numbers satisfying the equation $z^4 = iz$.

Which of the following statement(s) is/are correct?

(A) The complex number having least positive argument is $\left(\frac{\sqrt{3}}{2},\frac{1}{2}\right)$.

(B) $\sum_{k=1}^{3}\mathrm{Amp}(z_k)=\frac{\pi}{2}$

(C) Centroid of the triangle formed by z_1, z_2 and z_3 is $\left(\frac{1}{\sqrt{3}},\frac{-1}{3}\right)$

(D) Area of triangle formed by z_1, z_2 and z_3 is $\frac{3\sqrt{3}}{2}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 15 to 17)

For the complex number $w=\frac{4z-5i}{2z+1}$.

Q15. The locus of z, when w is a real number other than 2, is

(A) a point circle

(B) a straight line with slope $-\frac{5}{2}$ and y-intercept $\frac{5}{4}$

(C) a straight line with slope $\frac{5}{2}$ and y-intercept $\frac{5}{4}$

(D) a straight line passing through the origin

Q16. The locus of z, when w is a purely imaginary number is

(A) a circle with centre $\left(\frac{1}{2},-\frac{5}{4}\right)$ passing through origin.

(B) a circle with centre $\left(-\frac{1}{4},\frac{5}{8}\right)$ passing through origin.

(C) a circle with centre $\left(\frac{1}{4},-\frac{5}{8}\right)$ and radius $\frac{\sqrt{29}}{8}$

(D) any other circle

Q17. The locus of z, when $|w| = 1$ is

(A) a circle with centre $\left(-\frac{5}{8}, \frac{1}{4}\right)$ and radius $\frac{1}{2}$

(B) a circle with centre $\left(\frac{1}{4}, -\frac{5}{8}\right)$ and radius $\frac{1}{2}$

(C) a circle with centre $\left(\frac{5}{8}, -\frac{1}{4}\right)$ and radius $\frac{1}{2}$

(D) any other circle

Paragraph # 2 (Ques. 18 to 20)

Let A, B, C be three sets of complex numbers as defined below.

$A = \{z : |z+1| \le 2 + Re(z)\}$, $B = \{z : |z-1| \ge 1\}$

and $C = \left\{z : \left|\frac{z-1}{z+1}\right| \ge 1\right\}$

Q18. The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is

(A) 4 (B) 5

(C) 6 (D) 10

Q19. The area of region bounded by $A \cap B \cap C$ is

(A) $2\sqrt{3}$ (B) $\sqrt{3}$

(C) $4\sqrt{3}$ (D) 2

Q20. The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude & maximum modulus –

(A) –1 (B) $\frac{-3}{2}$

(C) $\frac{1}{2}$ (D) – 2

Paragraph # 3 (Ques. 21 to 24)

Suppose z and w be two complex numbers such that $|z| \le 1$, $|w| \le 1$ and $|z + iw| = 2$.

Use the result $|z|^2 = z\bar{z}$ and $|z + w| \le |z| + |w|$, answer the following questions.

Q21. Which of the following is true about $|z|$ and $|\omega|$

(A) $|z| = |w| = \frac{1}{2}$

(B) $|z| = \frac{1}{2}, |w| = \frac{3}{4}$

(C) $|z| = |w| = \frac{3}{4}$

(D) $|z| = |w| = 1$

Q22. Which of the following is true for z and ω

(A) $Re(z) = Re(\omega)$

(B) $I_m(z) = I_m(\omega)$

(C) $Re(z) = I_m(\omega)$

(D) $I_m(z) = Re(\omega)$

Q23. Number of complex numbers z satisfying the above conditions is

(A) 1 (B) 2

(C) 4 (D) indeterminate

Q24. Complex number w satisfying the above conditions is/are

(A) 1 or –i

(B) –1

(C) i or –1

(D) ω or ω^2 (ω is complex cube root of unit)

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q25. **Column-I** **Column-I**

(A) If z be the complex number such that $\left|z+\frac{1}{z}\right|=2$ (P) 0

then minimum value of $\frac{|z|}{\tan\frac{\pi}{8}}$ is

(B) $|z| = 1$ & $z^{2n}+1 \neq 0$ then $\frac{z^n}{z^{2n}+1}-\frac{\overline{z}^n}{\overline{z}^{2n}+1}$ is equal to (Q) 3

(C) If $8iz^3 + 12z^2 - 18z + 27i = 0$ then $2|z| =$ (R) 11

(D) If z_1, z_2, z_3, z_4 are the roots of equation (S) 1

$z^4 + z^3 + z^2 + z + 1 = 0$, then $\prod_{i=1}^{4}(z_i + 2)$ is

Q26. Match the equation in z, in **Column-I** with the corresponding values of arg(z) in **Column-II**.

Column-I (equations in z)	**Column-II** (principal value of arg (z))
(A) $z^2 - z + 1 = 0$	(P) $-2\pi/3$
(B) $z^2 + z + 1 = 0$	(Q) $-\pi/3$
(C) $2z^2 + 1 + i\sqrt{3} = 0$	(R) $\pi/3$
(D) $2z^2 + 1 - i\sqrt{3} = 0$	(S) $2\pi/3$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q27. If ω is the imaginary cube root of unity, then find the number of pairs of integers (a, b) such the $|a\omega + b| = 1$.

Q28. A polynomial f(z) when divided by $(z - \omega)$ leaves remainder $2+i\sqrt{3}$ and when divided by $(z - \omega^2)$ leaves remainder $2-i\sqrt{3}$. If the remainder obtained when f(z) is divided by $z^2 + z + 1$ is $az + b$ (where ω is non-real cube root of unity and $a, b \in R^+$), then find the value of $(a + b)$.

Q29. Let $A = \{a \in R|$ the equation $(1 + 2i)x^3 - 2(3 + i)x^2 + (5 - 4i)x + 2a^2 = 0\}$ has at least one real root. Find the value of $\sum_{a\in A} a^2$.

Q30. Complex number a, b and c zeros of the polynamial $P(z) = z^3 + qz + r$ (where $q, r \in R$) and $|a|^2 + |b|^2 + |c|^2 = 250$. If the points corresponding to a, b and c in the complex plane are the vertices of a right angled triangle with hypotenuse h, then $\left(\frac{h^2}{125}\right)$ is :

SUBJECTIVE TYPE

Q31. Find the modulus, argument and the principal argument of the complex numbers. $(\tan 1 - i)^2$

Q32. If one of the vertices of the square circumscribing the circle $|z - 1| = \sqrt{2}$ is $2+\sqrt{3}i$. Find the other vertices of square.

Q33. ABCD is a rhombus in the Argand plane. If the affixes of the vertices be z_1, z_2, z_3, z_4 and taken in anti-clockwise sense and $\angle CBA = \pi/3$, show that

(A) $2z_2 = z_1(1 + i\sqrt{3}) + z_3(1 - i\sqrt{3})$ &

(B) $2z_4 = z_1(1 - i\sqrt{3}) + z_3(1 + i\sqrt{3})$

Q34. If $iz^3 + z^2 - z + i = 0$, then prove that $|z|=1$.

Q35. P is a point on the Argand plane. On the circle with OP as diameter two points Q & R are taken such that $\angle POQ = \angle QOR = \theta$. If 'O' is the origin & P, Q & R are represented by the complex numbers Z_1, Z_2 & Z_3 respectively, show that :

$Z_2^2 \cos 2\theta = Z_1 . Z_3 \cos^2\theta$.

Q36. Find the sum of the series $1(2 - \omega)(2 - \omega^2) + 2(3 - \omega)(3 - \omega^2) \ldots\ldots (n - 1)(n - \omega)(n - \omega^2)$ where ω is one of the imaginary cube root of unity.

Q37. **(i)** Let Cr's denotes the combinatorial coefficients in the expansion of $(1 + x)^n$, $n \in N$.

If the integers

$a_n = C_0 + C_3 + C_6 + C_9 + \ldots\ldots$

$b_n = C_1 + C_4 + C_7 + C_{10} + \ldots\ldots$

and $c_n = C_2 + C_5 + C_8 + C_{11} + \ldots\ldots$,

then prove that

(a) $a_n^3 + b_n^3 + c_n^3 - 3a_n b_n c_n = 2^n$,

(b) $(a_n - b_n)^2 + (b_n - c_n)^2 + (c_n - a_n)^2 = 2.$

(ii) Prove the identity :

$(C_0 - C_2 + C_4 - C_6 + \ldots..)^2 + (C_1 - C_3 + C_5 - C_7 + \ldots\ldots)^2 = 2^n$

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
D	D	C	B	B	C	A	B	A	C
11	12	13	14	15	16	17	18	19	20
D	B	A	C	C	B	B	C	A	D
21	22	23	24	25	26	27	28	29	30
B	C	C	B	A	C	A	C	A	C
31	32	33	34	35	36	37	38	39	40
C	B	B	C	B	B	A	C	C	B
41	42	43	44	45	46				
A	D	A	C	D	B				

Ex - 1A

1. (B, C, D)	**2.** (A, B)	**3.** (A, D)	**4.** (B, C)	**5.** (A)
6. (C)	**7.** (D)	**8.** (A, B)	**9.** (A)	**10.** (A)
11. (A)	**12.** (B)	**13.** (B)	**14.** (A)	**15.** (C)

16. (A)

17. A $\rightarrow$ Q; B $\rightarrow$ P; C $\rightarrow$ Q, S; D $\rightarrow$ R

18. A $\rightarrow$ P, T; B $\rightarrow$ Q, R, S; C $\rightarrow$ S, T

19. 2 **20.** 1 **21.** 51 **22.** 136 **23.** -160

24. (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $-\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

25. $\arg z = \frac{\alpha}{2}+\frac{\pi}{4}-\pi = \frac{\alpha}{2}-\frac{3\pi}{4}$, $|z| = \sqrt{2}\left(\sin\frac{\alpha}{2}-\cos\frac{\alpha}{2}\right)$

26. $|z| = 1$, $\text{amp}(z) = -\frac{\pi}{3}$

27. $z_p = 20\left(\frac{3}{5}+i\frac{4}{5}\right)$; $z_p = 12+16i$; $z_E = 10i$; $z_D = 40i$

28. (i) Polar form is $z = \sqrt{2}\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$; exponential form is $z = \sqrt{2}\,e^{\frac{3\pi}{4}i}$

(ii) Polar form is $z = \sqrt{2}\left(\cos\frac{5\pi}{12}+i\sin\frac{5\pi}{12}\right)$; exponential form is $z = \sqrt{2}e^{\frac{5\pi}{12}i}$

29. $3\omega^2$

30. $z_3 = z_1 + (z_2 - z_1)(1 + i)$; $z_4 = z_2 + (1 + i)(z_1 - z_2)$

32. (A) $x^2 - 2x + 1 + y^2 + x^2 + 2x + 1 + y^2 = 4 \Rightarrow x^2 + y^2 = 1$

(B) $y = \pm x$

Ex-2

1	2	3	4	5	6	7	8	9	10
C	D	B	D	D	D	D	A	B	C
11	12	13	14	15	16	17	18	19	20
C	C	C	D	A	B	B	A	C	C
21	22	23	24	25	26	27	28	29	
C	D	D	D	D	D	A	B	B	

Ex - 2A

1. A, B, C **2.** B, C **3.** A,B,C,D **4.** (A,C) **5.** B,C,D

6. A,B,D **7.** A,C,D **8.** A,C,D **9.** B, C, D **10.** D

11. A **12.** A,B,D **13.** A, B, D **14.** A,B **15.** C

16. B **17.** D **18.** C **19.** A **20.** B

21. D **22.** D **23.** B **24.** C

25. A $\rightarrow$ S; B $\rightarrow$ P; C $\rightarrow$ Q; D $\rightarrow$ R

26. A $\rightarrow$ Q, R; B $\rightarrow$ P, S; C $\rightarrow$ Q, S; D $\rightarrow$ P, R

27. 6 **28.** 5 **29.** 18 **30.** 3

31. Modulus = $\sec^2 1$, Arg $z = 2n\pi + (2 - \pi)$, Principal Arg $z = (2 - \pi)$

32. $(-\sqrt{3}\,i), (1-\sqrt{3})+i$ and $(1+\sqrt{3})-i$ **36.** $\left[\frac{n(n+1)}{2}\right]^2 - n$

-: Important Notes & Formulas :-